Exercise sheet: Week 2 - Solutions

1. (a) By using the fact that the sample is i.i.d, we find that the likelihood is given by

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}$$

and so $l(\lambda) = n \log \lambda - \lambda \sum x_i$ and

$$l'(\lambda) = \frac{n}{\lambda} - \sum x_i = 0\,,$$

which gives $\widehat{\lambda} = \frac{1}{\overline{x}}$.

We check this is a maximum: $l''(\lambda) = -\frac{n}{\lambda^2} < 0$.

(b)

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta} = \theta^{-n} e^{-\frac{1}{\theta}\sum x_i}$$

and so $l(\theta) = -n \log \theta - \frac{1}{\theta} \sum x_i$ and

$$l'(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum x_i = 0\,,$$

which gives $\widehat{\theta} = \frac{\sum x_i}{n} = \overline{x}$.

We check this is a maximum: $l''(\theta) = \frac{n}{\theta^2} - \frac{2n\overline{x}}{\theta^3}$, which is < 0 when $\theta = \widehat{\theta} = \overline{x}$.

Note that $\theta = \frac{1}{\lambda}$ and $\widehat{\theta} = \frac{1}{\widehat{\lambda}}$, illustrating the invariance property of the MLE. See the lecture notes.

2.

$$L(\alpha) = \prod_{i=1}^{n} \alpha x_i^{\alpha-1} = \alpha^n \left(\prod x_i\right)^{\alpha-1}$$

$$\Rightarrow l(\alpha) = n \log \alpha + \alpha \sum \log(x_i) + \text{constant}$$

$$\Rightarrow l'(\alpha) = \frac{n}{\alpha} + \sum \log(x_i) = 0$$

which gives $\widehat{\alpha} = \frac{-n}{\sum \log(x_i)}$.

We check this is a maximum: $l''(\alpha) = -\frac{n}{\alpha^2} < 0$.

We recall now that $x \in (0, 1)$. It can be observed that $\widehat{\alpha}$ will be large when $x_i \to 1$, that is, $\log(x_i) \to 0$, and so $\widehat{\alpha} \to +\infty$.

To find the MME, note that if $X \sim \text{Beta}(\alpha, 1)$ then $\text{E}(X) = \frac{\alpha}{\alpha+1}$. Hence, the method of moments estimator is

$$\widetilde{\alpha} = \frac{\overline{x}}{1 - \overline{x}}\,.$$

In general, this is different from the MLE.

3. (a) We use the factorisation theorem:

$$\begin{aligned} f_{\underline{X}}(\underline{x};\beta) &= \left(\frac{\beta^2}{\Gamma(2)}\right)^n \prod_{i=1}^n x_i e^{-\beta x_i} \\ &= \beta^{2n}\left(\prod x_i\right) e^{-\beta\sum x_i} \\ &= \left(\prod x_i\right) \times \beta^{2n} e^{-\beta\sum x_i} \\ &= h_1(\underline{x}) \times h_2(\beta, \sum x_i)\,. \end{aligned}$$

Hence, $\sum X_i$ is sufficient for β.

(b) Using the factorisation theorem:

$$\begin{aligned} f_{\underline{X}}(\underline{x};\alpha) &= \left(\frac{\Gamma(\alpha+1)}{\Gamma(\alpha)}\right)^n \prod_{i=1}^n (1-x_i)^{\alpha-1} \\ &= \alpha^n \left[\prod (1-x_i)\right]^{\alpha-1} \\ &= 1 \times \alpha^n \left[\prod (1-x_i)\right]^{\alpha-1} \\ &= h_1(\underline{x}) \times h_2\left(\alpha, \prod (1-x_i)\right)\,. \end{aligned}$$

Hence, $\prod(1-X_i)$ is sufficient for α.

4. (a) Observations $x_1, \ldots, x_n$ are a random sample from the Pareto distribution with pdf $f_X(x;\theta) = \frac{\theta}{(1+x)^{\theta+1}}$ for $0 < x < \infty$ and $\theta > 0$. Hence

$$\begin{aligned} L(\theta) = f_{\underline{X}}(\underline{x};\theta) &= \prod_{i=1}^n \frac{\theta}{(1+x_i)^{\theta+1}} \\ &= \theta^n \left[\prod (1+x_i)\right]^{-\theta-1} \\ &= \theta^n \exp\left\{-(\theta+1)\sum \log(1+x_i)\right\} \\ &= h\left(\theta, \sum \log(1+x_i)\right)\,. \end{aligned}$$

Hence, by the factorisation theorem, $\sum \log(1+X_i)$ is sufficient for θ.

Now,

$$\begin{aligned} l(\theta) &= n\log\theta - (\theta+1)\sum \log(1+x_i) \\ \Rightarrow l'(\theta) &= \frac{n}{\theta} - \sum \log(1+x_i) = 0\,. \end{aligned}$$

Solving this gives the MLE $\widehat{\theta} = \frac{n}{\sum \log(1+X_i)}$.

We check that this is a maximum: $l''(\theta) = -\frac{n}{\theta^2} < 0$.

(b) Let $Y_i = \log(1 + X_i)$.

Note that if $y = \log(1 + x)$ then $0 < y < \infty$ and $x = e^y - 1$. Hence $\frac{dx}{dy} = e^y$. Therefore

$$\begin{aligned} f_Y(y) &= f_X(x) \left| \frac{dx}{dy} \right| \\ &= \theta \left(1 + e^y - 1\right)^{-(\theta+1)} e^y \\ &= \theta e^{-\theta y} \\ &= \frac{1}{\mu} e^{-\frac{y}{\mu}}, \end{aligned}$$

for $y > 0$, where $\mu = \frac{1}{\theta}$.

Hence, $Y_i \sim \text{Exp}(1/\mu)$, as required. Therefore (as in Q1)

$$\widehat{\mu} = \overline{Y} = \frac{1}{n} \sum \log(1 + X_i) = \frac{1}{\widehat{\theta}}.$$

This illustrates the invariance property of the MLE.

5. (a)

$$L(p) = \binom{n_1}{x_1} p^{x_1} (1-p)^{n_1 - x_1} \times \binom{n_2}{x_2} p^{x_2} (1-p)^{n_2 - x_2}$$

$$\Rightarrow l(p) = \text{c} + (x_1 + x_2) \log(p) + (n_1 + n_2 - x_1 - x_2) \log(1-p)$$

$$\Rightarrow l'(p) = \frac{x_1 + x_2}{p} - \frac{n_1 + n_2 - x_1 - x_2}{1-p} = 0$$

Solving this we get $(x_1 + x_2)(1-p) = (n_1 + n_2 - x_1 - x_2)p$, and so

$$\widehat{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

(b) In this case

$$L(p_1, p_2) = \underbrace{\binom{n_1}{x_1} p_1^{x_1} (1-p_1)^{n_1 - x_1}}_{\text{(A)}} \times \underbrace{\binom{n_2}{x_2} p_2^{x_2} (1-p_2)^{n_2 - x_2}}_{\text{(B)}}$$

$$\Rightarrow l(p_1, p_2) = \text{c} + x_1 \log(p_1) + (n_1 - x_1) \log(1-p_1) + x_2 \log(p_2) + (n_1 - x_2) \log(1-p_2) .$$

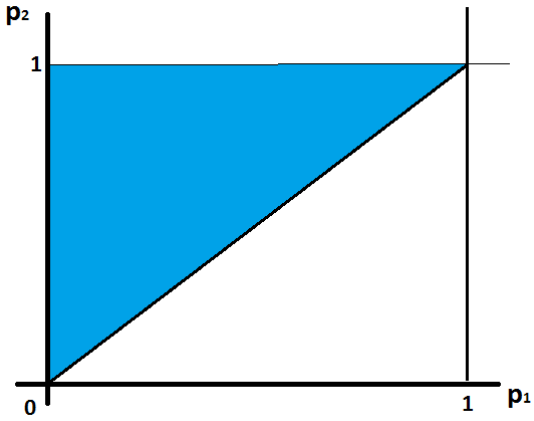
$$l'(p_1) = \frac{x_1}{p_1} - \frac{n_1 - x_1}{1-p_1} = 0$$

$$l'(p_2) = \frac{x_2}{p_2} - \frac{n_2 - x_2}{1-p_2} = 0$$

It can be observed that (A) is maximised by $\widehat{p_1} = \frac{x_1}{n_1}$ and (B) is maximised by $\widehat{p_2} = \frac{x_2}{n_2}$.

Therefore, $(\widehat{p_1}, \widehat{p_2}) = \left(\frac{x_1}{n_1}, \frac{x_2}{n_2}\right)$ is the MLE.

(c) If we force $p_2 \geq p_1$, then the parameter space for (p_1, p_2) corresponds to the shaded triangle below:



Case 1: If $\frac{x_1}{n_1} > \frac{x_2}{n_2}$ then the solution $(\widehat{p_1}, \widehat{p_2})$ which maximises $L(p_1, p_2)$ over the whole square will be *outside* the shaded triangle. In this case, the value inside the shaded triangle that maximises $L(p_1, p_2)$ will lie on the boundary line $p_1 = p_2$.

If we maximise $L(p_1, p_2)$ subject to the condition that $p_1 = p_2$ then we obtain $(\widehat{p_1}, \widehat{p_2}) = \left(\frac{x_1+x_2}{n_1+n_2}, \frac{x_1+x_2}{n_1+n_2}\right)$, as in part (a).

Case 2: If $\frac{x_1}{n_1} \leq \frac{x_2}{n_2}$, then $(\widehat{p_1}, \widehat{p_2}) = \left(\frac{x_1}{n_1}, \frac{x_2}{n_2}\right)$ lies inside the shaded region, and is therefore the MLE.

The situation $p_2 \geq p_1$ could arise if the second experiment involved a test which was more sensitive than that used in the first experiment.

Example: A genetic mutation causes a physical characteristic in a carrier of the mutation with probability $q \leq 1$. If p_2 represents the proportion of the population that carries the gene, then $p_1 = p_2 q$ represents the proportion with the physical characteristic.

Suppose our first experiment selects n_1 individuals and counts the number x_1 with the physical characteristic, while the second experiment selects n_2 individuals and records how many carry the genetic mutation (for example, using DNA analysis). In this situation we have $p_1 \leq p_2$.